

Abhinav Gyawali

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Education

University of Texas at Austin
PhD Computer Science
Advisors Dixin Tang, Vijay Chidambaram

Aug 2026 - Now

Tribhuvan University
BS Computer Science and Information Technology

Oct 2019 – Aug 2024

Publications

Mihai Budiu, Leonid Ryzhyk, Gerd Zellweger, Ben Pfaff, Lalith Suresh, Simon Kassing, **Abhinav Gyawali**, Matei Budiu, Tej Chajed, Frank McSherry, Val Tannen. **DBSP: Automatic Incremental View Maintenance for Rich Query Language**. *The International Journal on Very Large Data Bases (VLDB)*

May 2025

Abhinav Gyawali. **SocketDB: DBMS with Data Streaming via WebSockets**. *Deerwalk Journal of Computer Science and Information Technology (DJCSIT)*

Sep 2024

Experience

Feldera, Software Engineer

US, Remote

Dec 2023 – July 2026

- Enhanced Feldera's SQL engine by extending compiler support for complex types like **VARIANT**, while also building automated testing infrastructure using **SQLancer** [🔗](#), uncovering **31 previously undetected bugs**.
- Designed and implemented distributed systems components including real-time output connectors (**Postgres**, **Redis**), S3-backed checkpointing and recovery, and fault-tolerance mechanisms for pipeline execution.
- Developed the **Feldera Python SDK** [🔗](#) for programmatic pipeline control and authored technical documentation and use-case guides for stream processing, batch processing, and observability.

Invisid, Software Developer

Germany, Remote

May 2022 – Nov 2023

- Developed **trackpad-rs** [🔗](#), a Rust library for extracting high-fidelity trackpad and mouse interaction data from Apple's *MultitouchSupport* framework for continuous biometric authentication and rapid attacker detection.
- Built macOS client software for continuous behavioral authentication and re-designed backend server infrastructure to improve efficiency, scalability, and consistency of biometric data processing.

Flow Webinar, Security Engineer (Part time)

US, Remote

May 2020 – June 2021

- Conducted vulnerability assessment and penetration testing of a WebRTC-based video conferencing platform, while also managing backend infrastructure and system administration for production deployments.

Projects

DBSP

[🔗 feldera/feldera/dbsp](#)

- DBSP is an incremental computation engine where changes to the data set run in time proportional to the size of the change rather than the size of the data set.

SocketDB

 [abhizer/socketdb](#)

- SocketDB is a lightweight SQL database that enables real-time updates through WebSockets.

Nyx-lang

 [abhizer/nyx-lang](#)

- Nyx-lang is a statically typed, tree-walking interpreted language with a type-checking mechanism to ensure safety and correctness.

Loogle-rs

 [abhizer/loogle-rs](#)

- Loogle is a "Local Google" like search engine based on Term Frequency - Inverse Document Frequency (TF-IDF).

Monkey-rs

 [abhizer/monkey-rs](#)

- Monkey-rs is my take on the concepts from the [Writing an Interpreter in Go](#) book.

Awards & Certifications

UTCS Distinguished Graduate Fellowship	2026
SIGPLAN-M Mentee (Mentor: Professor Jonathan Balkind)	2025
DWIT Merit Based Scholarship	2021
DWIT Merit Based Scholarship	2020
AWS Educate Student Ambassador	2020
Red Hat Certified Engineer (170-197-891)	2017
Red Hat Certified System Administrator (170-197-891)	2017